BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2017

CHEMISTRY

Past Paper 2017

Retyped faithfully from the original board paper — same questions, no changes.

Section-A — Multiple Choice Question (MCQs) — Q No: 01: Choose the Correct Answer.

Q(i). Metals which have low melting and boiling points belongs to group.

(A) IV A
(B) III A
(C) II A
(D) I A

Q(ii). Formula of Quick lime is:

(A) $Ca(OH)_2$
(B) CaO
(C) NaOH
(D) Na_2O

Q(iii). The ion which is smaller than parent atom is known as:

(A) Negative ion
(B) Positive ion
(C) Neutral ion
(D) None

Q(iv). Tincal is the name of:

(A) Alunite
(B) Diaspore
(C) Borax
(D) Gibbisite

Q(v). Benzene was first isolated from coal tar in the year:

(A) 1845
(B) 1820
(C) 1825
(D) 1835

Q(vi). SN^2 reaction occurs in steps:

(A) Three
(B) Two
(C) One
(D) None

Q(vii). The boiling point of methanol is:

(A) 63.7°C
(B) 64.7°C
(C) 70°C
(D) 80°C

Q(viii). The visual pigment is formed by the vitamin:

(A) D
(B) C
(C) B
(D) A

Q(ix). Chromo proteins are coloured due to presence of the group:

(A) Glyco – protein
(B) Lipo – protein
(C) Non –protein
(D) Phospho – protein

Q(x). Radio – active isotope of hydrogen is _____

(A) Protium
(B) Deutrium
(C) Tritium
(D) All of these

Q(xi). In periodic table the first long period is:

(A) Period – I
(B) Period – 2
(C) Period – 3
(D) Period – 4

Q(xii). The formula of Phosphine is:

(A) PH_2
(B) PH_3
(C) PH_4
(D) PO_3

Q(xiii). The hydrides which are intermediate between metallic and covalent hydrides are known as:

(A) Complex
(B) Polemeric
(C) Border line
(D) Ionic

Q(xiv). Bras is an alloy of:

(A) Fe-Zn
(B) Cu-Sn
(C) Cu-Zn
(D) Pb-Fe

Q(xv). Blister is the name of impure:

(A) Sodium
(B) Iron
(C) Zinc
(D) Copper

Q(xvi). The I.U.P.A.C. name of iso – propyl alcohol is:

(A) 2 – propanol
(B) 1 – propanol
(C) 3 – Propanol
(D) 2-propanal

Q(xvii). Which of the following hydrocarbon on inhalation produces anacesthesia.

(A) Methane
(B) Ethene
(C) Propene
(D) Butene

SECTION – B

Q2. State the periodic Law. In what various ways it help the chemist?

Q3. Complete the following reaction. (i) $PCl_3 + H_2O \rightarrow$ (ii) $Mg_3N_2 + H_2O \rightarrow$ (iii) $Ca_3P_2 + H_2O \rightarrow$ (iv) $Al_4C_3 + H_2O \rightarrow$ (v) $CaC_2 + H_2O \rightarrow$

Q4. How aluminium is extracted from bauxite by Hall's and Serpek's method?

Q5. Prove that HNO_3 is a strong Oxidant.

Q6. Discuss why water and ammonia molecules act as ligands but H_3O^+ and NH_4^+ do not?

Q7. What is meant by: (i) Catenation (ii) Homologous series (iii) Isomerism (iv) Cracking (v) Polymerization

Q8. Write electronic structure of the following molecules. (i) Acetone (ii) Nitroethane (iii) Phenlmagnesium bromide (iv) Silver acetylide (v) Mono sodium acetylide

Q9. What is meant by hydrophobic and hydrophilic property of detergents?

Q10. Discuss importance and structure of Tocopherol.

Q11. Describe briefly elements of Nutrition.

SECTION – C

Q12.

(a) Discuss the importance of the following. (i) Cl_2 (ii) HNO_3 (iii) H_2SO_4

(b) Describe the extraction of sodium from common salt. How and under what conditions sodium reacts with: (i) Water (ii) Hydrogen (iii) Nitrogen (iv) Oxygen

Q13.

(a) Write note on any two of the following. (i) Physical and chemical properties of acetic acid (ii) Reactions of benzene (iii) Metallurgy of metals

(b) What are amino acids? How can an amino acid act as buffer?

Q14.

(a) Write structure of the following. (i) Ferrocyanide (ii) Diethylene triamine (iii) Acetophenone (iv) sodium benzoate (v) Methyl formate (vi) Aspargine (vii) Zwitterion

(b) Name the following compounds [structure diagrams in source scan]: (i) $CH_3COO(CH_3)CH_3$ (ii) $CH_3–CH–COO–$ with OH and CH_3 groups (iii) $[Ag(NH_3)_2]OH$ (iv) $CH_3–CH–O–CH_3$ with a CH_3 branch (v) $[Pt(NH_3)_2Cl_4]$ (vi) $[Cu(NH_3)_4]SO_4$ (vi) $CH_3–C(CH_3)–CH_2–CO–C_2H_5$